FRA WindLab | Model card

AIW-AIRWAYS / Direction v2 + closure v1 / 30 October 2026 / Submission partial

Research prototype: 48 hourly forecasts of RWY18 wind usability and OD07/25, direction scores and first predicted change hour. Direction v2 learns from DFLD-reported operations; RWY18 retains weather-rule proxy targets. Scores are not calibrated operational certainty. Intermediate leads reuse the nearest 1/6/24/48h model; change timing is not independently validated.

Held-out DFLD direction quality

Lead	Test hours	New accuracy	Old / rules accuracy	New Brier	Old / rules Brier
1h	517	79.69%	76.98% / 76.21%	0.1900	0.2186 / 0.2379
6h	517	80.08%	75.63% / 76.21%	0.1965	0.2088 / 0.2379
24h	521	71.79%	71.79% / 71.02%	0.2371	0.2317 / 0.2898
48h	521	70.06%	69.48% / 68.52%	0.2415	0.2308 / 0.3148

Identical held-out DFLD labels for all three methods. Lower Brier is better. New accuracy gains vs original: +2.71pp / +4.45pp / 0pp / +0.58pp. New probability scores improve at 1h/6h but regress at 24h/48h. Weather rules are beaten at every lead. These scores are not comparable with the older weather-proxy benchmark.

Data and clock alignment

183 days, 30 April-29 October 2026: 8,784 DFLD half-hour direction reports. DFLD describes Frankfurt direction since March 2004 as Fraport-based. Europe/Berlin is empirically supported by 240/240 exact wind matches against UTC METAR across five summer/winter dates, not an explicit provider declaration. The ambiguous 25 October DST date is excluded from UTC model joins. Original METAR wind is used; variable/missing direction is excluded. Runway counts exist for 27-29 October only; no-use never implies closure.

Learning, drivers and chronology

Independent L2 logistic direction classifiers use archived ICON wind, runway components, origin METAR, past 3h trends, UTC hour and nonlinear wind interactions. UTC target splits: train 30 Apr-31 Aug; validate 4-26 Sep; test 30 Sep-29 Oct 2026, with 72h gaps. Train-only scaling; feature set and regularization selected on validation Brier. No test tuning. Test coverage is about 75% after excluding missing/variable inputs. DFLD labels are targets only, never forecast-time inputs.

Forecast-time and uncertainty limits

49,600 EDDF METAR reports from IEM and DWD ICON archives via Open-Meteo support weather features. Origin observations are at most 40min old; trends use past data. Nominal forecast offsets predate origin by at least 18h, but exact delivery timestamps are unknown. Live ICON is fresher than backtest inputs. Six months have limited seasonality; daily-block bootstrap intervals are in the dashboard. Retrospective replay is not an as-issued forecast.

RWY18 and airport hours

Closure model and 1% / 10% alert thresholds are unchanged. At 50%, it detects 0/5 all-hour closure proxies. At 1h during scheduled hours, catch mode detects 3/3 with 116 false alarms; quiet mode 0/3 with 4 false alarms. Both override on forecast wind limits. No actual closure truth was obtained. Scheduled flights: 05:00-23:00 Europe/Berlin; night restrictions are separate from wind. Conditional delays and exemptions do not imply individual permission.

Challenge limits and remaining submission

RWY18 sustained tailwind >=15 kt or gust crosswind >40 kt restricts use. Direction switches with tailwind >=10 kt or smaller adverse wind >30min; challenge headings 180/070/250 degrees. Public Fraport landing guidance differs (5 kt). Jury dates, prospective weather timestamps, confirmed closure labels and switch-event scoring remain outstanding. Replay: model/predict.py after 27 Sep 2026 selection; earlier replay requires --legacy-direction.

Sources and reproduction

DFLD history | DFLD methodology | IEM METAR | ICON archives | Fraport hours | Code and full provenance

Full URLs, retrieval times, hashes, model coefficients, paired comparisons and prediction rows are downloadable from the dashboard. Reproduce direction fitting: model/train_dfld.py; audit: tests/direction-model.mjs. DFLD raw pages and aircraft identifiers are not redistributed. Numeric challenge rules: supplied FraAlliance case PDF.